

John Marsden

john@johnmarsden.dev | +1 (325) 386-8728 | Raleigh, North Carolina, US

[LinkedIn](#) | [GitHub \(Professional\)](#) | [GitHub \(Personal\)](#)

EDUCATION	<u>North Carolina State University</u>	Graduation Date: 04/2025
	M.S. – Computer Science	
	<u>Abilene Christian University</u>	Graduation Date: 04/2020
	B.S. – Math and Computer Science	
SKILLS	<u>Teaching:</u> Curriculum Development, Lesson Planning, Student Assessment, Flipped Classroom, Universal Design for Learning <u>Instructional Technology:</u> Learning Management Systems, Canvas, Moodle <u>Programming Languages:</u> C#, Python, Java, SQL, Rust, HTML, CSS, JavaScript, F#, C++, C, Haskell <u>Dev Ops:</u> Fuzzing, Github Actions, Docker, Bash, Linux	
EXPERIENCE	<u>Teaching Assistant</u> NCSU	07/2020 – 12/2024
	- Helped 300+ students navigate the complexities of multi-project Java codebases - Validated 700+ student algorithm analyses and algorithm designs - Graded 400+ student algorithms in GradeScope using common data structures, including stacks, graphs, etc.	
	<u>PhD Student</u> AIAL	05/2021 – 11/2024
	- Implemented a novel intermediate language for representing student code for use by graph-based neural networks - Performed a comprehensive systematic literature review of 500+ papers from the debugging education literature - Designed 3 studies for use by our research lab to investigate a variety of methods for better teaching and learning debugging	
	<u>Instructor - Software Development Fundamentals</u> NCSU	05/2022 – 07/2022
	- Taught the software development process and test-driven development to 20 students - Engaged students through question and answer sections in each lecture - Created and administered three exams to validate students' learning progress	
	<u>Instructor - Introduction to Computing</u> NCSU	12/2021 – 04/2022
	- Taught the fundamentals of Java to 40 students - Built two autograders using GradeScope and JUnit 5 to autograde student code submissions - Designed and administered three exams to validate students' learning progress - Used the Moodle LMS course page to share lecture slides, communicate assignments, and provide resources	
	<u>Software Development Intern</u> Secmation	04/2021 – 07/2021
	- Planned and implemented 20% of the build system for a secure UAV programming environment with Python - Designed an intuitive API for the backend using AsyncAPI - Containerized the programming environment backend using Docker for scalability and ease of use	
	<u>Teaching Assistant</u> ACU	07/2017 – 04/2020
	- Tutored 70+ students on C++ programming concepts, including the basics, pointers, memory management, and object-oriented programming - Graded 200+ student C++ algorithm implementations using common data structures, including queues, graphs, trees, etc.	
	<u>Software Development Intern</u> Faithlife	04/2019 – 07/2019
	- Converted the OAuth application management portal from being user specific to a group-based system - Integrated a HavelBeenPwned check using the k-anonymity API - Collaborated with a team of 4 to stand up a new Varnish cache server for the Logos internal Hackathon	